

COMSATS UNIVERSITY ISLAMABAD
(ATTOCK Campus)

NAME

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REG. No

SP22-BSE-055

Dated

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(ASSIGNMENT-2)

Q1 Find the transition probability P_{ij} .

Solution

Count transition from each state.

⇒ Transition from state 0

$$0 \rightarrow 0 = 8$$

$$0 \rightarrow 1 = 12$$

$$0 \rightarrow 2 = 12$$

$$\text{Total} = 32$$

⇒ Transition from state 1

$$1 \rightarrow 0 = 5$$

$$1 \rightarrow 1 = 2$$

$$1 \rightarrow 2 = 16$$

$$\text{Total} = 23$$

⇒ Transition from state 2

$$2 \rightarrow 0 = 2$$

$$2 \rightarrow 1 = 2$$

$$2 \rightarrow 2 = 0$$

$$\text{Total} = 4$$

$$\text{Total transitions} = 32 + 23 + 4 = 59$$

Transitions Probabilities

⇒ For State 0:

$$P_{00} = 8/32 = 0.25$$

$$P_{01} = 12/32 = 0.375$$

$$P_{02} = 12/32 = 0.375$$

⇒ For State 1:

$$P_{10} = 5/23 = 0.217$$

$$P_{11} = 2/23 = 0.086$$

$$P_{12} = 16/23 = 0.695$$

⇒ For State 2:

$$P_{20} = 2/4 = 0.5$$

$$P_{21} = 2/4 = 0.5$$

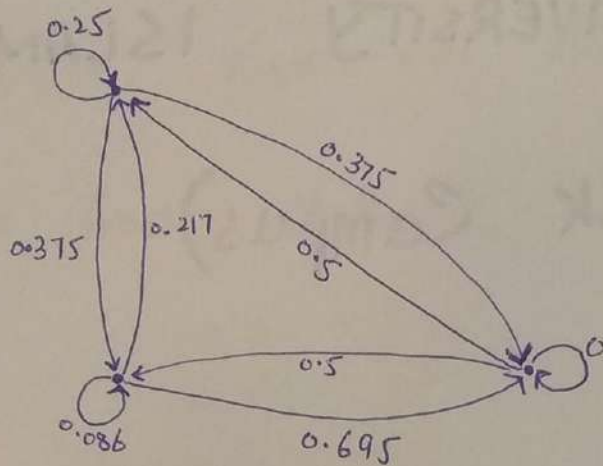
$$P_{22} = 0/4 = 0$$

(b) Mathematical Model

Sample space = $\{0, 1, 2\}$

$$P = \begin{bmatrix} P_{00} & P_{01} & P_{02} \\ P_{10} & P_{11} & P_{12} \\ P_{20} & P_{21} & P_{22} \end{bmatrix} = \begin{bmatrix} 8/32 & 12/32 & 12/32 \\ 5/23 & 2/23 & 16/23 \\ 2/4 & 2/4 & 0/4 \end{bmatrix}$$

c) State transition diagram



(d) Transition Probability Matrix

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 0.25 & 0.375 & 0.375 \\ 0.217 & 0.086 & 0.695 \\ 0.5 & 0.5 & 0 \end{bmatrix} \end{matrix}$$

e) What is P^4_{01}

$$P^4 = P \times P \times P \times P$$

$$P^4 = \begin{bmatrix} 0.407 & 0.323 & 0.270 \\ 0.392 & 0.316 & 0.292 \\ 0.431 & 0.332 & 0.237 \end{bmatrix}$$

$$\boxed{P^4_{01} = 0.323}$$